Robotic Car Painting Cell in RoboDK

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Software: RoboDK 6.1 | Robot: ABB IRB 5500-27 on ABB IRBT 7004 track
GitHub: https://github.com/1tsMS/paint-booth-sim

1. Summary

A car body rides in on a conveyor, stops in the booth and an ABB IRB 5500-27 paint robot on a linear track paints the hood, front fender, door, rear fender, trunk and roof. The car then leaves through the exit. All six panel programs pass RoboDK's path check with collision checking on. One side of the car takes about 181 s to paint, and the full cell cycle with conveyor moves takes about 225 s.

2. Software used

Software	Used for
RoboDK 6.1	Cell simulation, offline programming, path and collision checks, spray simulation
Fusion 360	Booth walls, conveyor rails, supports and cart
Blender	Cleaning and scaling the car body mesh

I also considered Gazebo/ROS and MuJoCo. They are good physics simulators, but they don't come with industrial paint robots, linear tracks or offline programming tools, so most of the time would have gone into setup. RoboDK has all of these and is used for offline programming in industry.

3. Robot selection and reach study

The car is 4.8 m long, so the first question was how many robots and where to put them. I drew the car cross-section to scale next to an IRB 5500 arm. One fixed arm reaches the near side and the roof centre, but it can't reach the lower panels on the far side without its links hitting the body.

- **Robot:** ABB IRB 5500-27, a dedicated paint robot (6 axes, 2.2 m reach in the RoboDK model). I started with the wall-mounted -23 version and moved to the -27 because its base mounts directly on a track carriage.
- **Track:** ABB IRBT 7004 with 3700 mm travel, synchronised as a 7th axis, so a 2.2 m arm can cover the full car length.
- **Robot count:** a real line would use two mirrored robots, one per side. The brief asks for one robot, so I programmed one side.

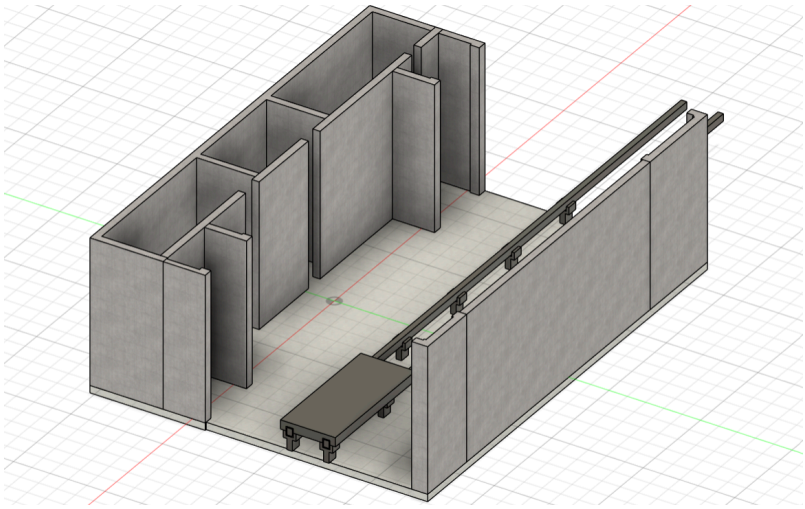
Alternatives I considered:

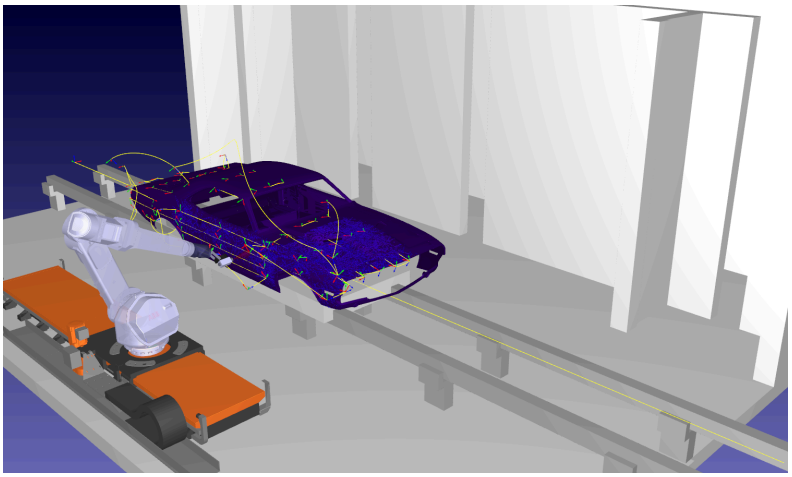
- Scanning the car with LiDAR and planning paths from the scan: dropped, because a plant already has the CAD of every model. Offline programming from known geometry is the standard approach.
- An overhead XYZ gantry with a tilting head: dropped, because it is a custom machine

rather than an industrial robot, and moving parts over a wet car disturb the booth downdraft and drop particles on the paint.
- One robot with a turntable for the car: possible, but a 4.8 m car needs a 5.16 m turning circle.

4. Cell layout

The reference layout's dimensions don't add up (the three top rooms total 9500 mm against the 12000 mm overall length), so I used it as a guide and drew my own layout in Fusion 360 around the robot track and conveyor.





Element	Size / position
Full booth	12200 x 9200 mm, walls 4000 mm high
Main paint booth	7000 x 8000 mm
Entry / exit zones	5610.9 mm
Conveyor rails	2 rails, 200 x 150 mm, 17200 mm long
Conveyor cart	3500 x 1500 mm, inside the car footprint so it stays out of the spray
Conveyor stops	Entry 0 mm, paint stop 4500 mm, exit 13248 mm, at 300 mm/s
Robot track	Left side of the car, robot base 2496 mm from the car centreline

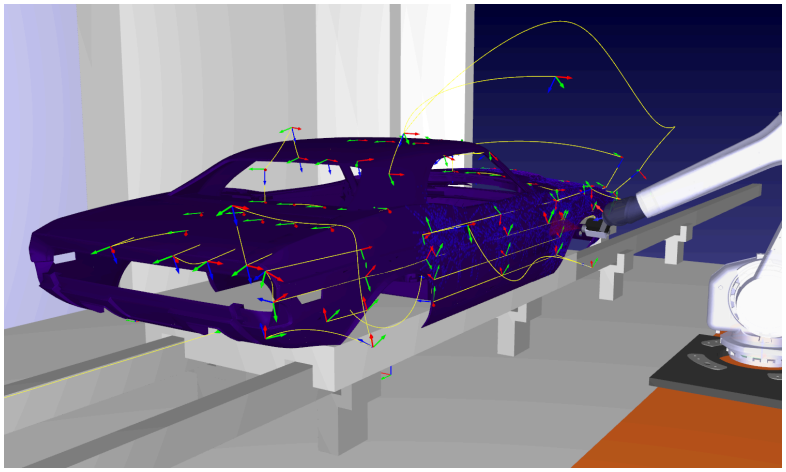
Car body: a free Sketchfab model ("dodge car body", CC-BY-4.0). In Blender I fixed its scene transforms, set the origin at the bottom centre, scaled it uniformly to 4801 x 1962 x 1115 mm and reduced it from 433k to 132k triangles. Uniform scaling keeps the real proportions, so it is lower than the 1600 mm reference height.
Tool calibration: I used RoboDK's generic paint sprayer. Since the targets sit on the car surface, the TCP has to be the standoff point in front of the nozzle. Measuring the nozzle face and the TCP in the same frame showed the TCP was only 174 mm in front of the nozzle, so I moved it along the tilted spray axis to 250 mm: TCP = [87.9, 0, 515.6] mm, Ry = 30 deg.

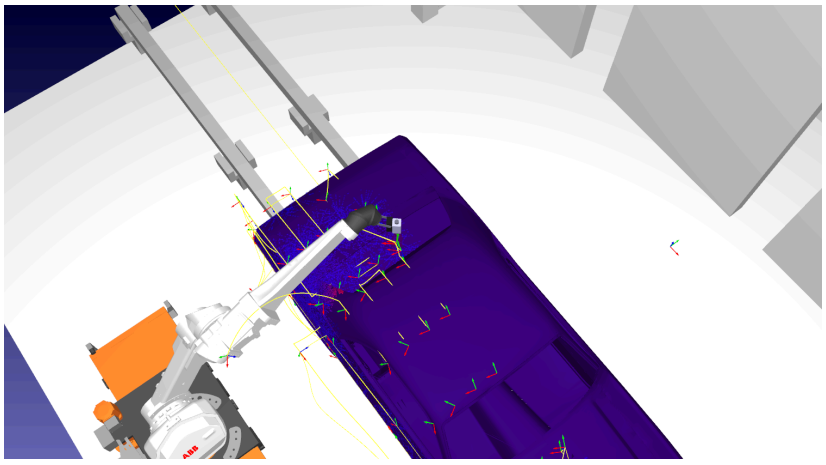
5. Trajectory and path planning

Parameter	Value
Standoff	250 mm (built into the TCP)
Spray width	300 mm
Pass spacing	150 to 225 mm, last pass runs past the centreline for overlap with the second robot
Speed	500 mm/s painting, 1000 mm/s approach
Edges	Strokes start about 100 mm off a free panel edge

- Each panel is a set of zig-zag passes with 3 targets per pass (start, middle, end), taught in the car frame, which rides on the conveyor.
- Every target sits on the panel surface with the gun along the surface normal.
- The track moves with the gun so the robot base stays roughly opposite the target, and the arm keeps a similar pose along a pass.
- Linear moves along the strokes and joint moves between passes. A few strokes where a straight line wasn't reachable run as joint moves.
- The spray turns on at the start of each pass and off at its end, so it is off during approaches and panel changes.

Program	Targets	Layout	Time
Paint_Hood	15	5 passes, front to rear	26.4 s
Paint_Fender	8	Upper pass, then one stroke around the wheel arch	16.5 s
Paint_Door	12	4 horizontal passes	32.0 s
Paint_Rear_Fender	9	2 passes and a low pass behind the arch	30.6 s
Paint_Trunk	15	5 passes, starts at the corner next to the rear fender	23.1 s
Paint_Roof	12	4 passes, starts at the corner next to the trunk	33.6 s





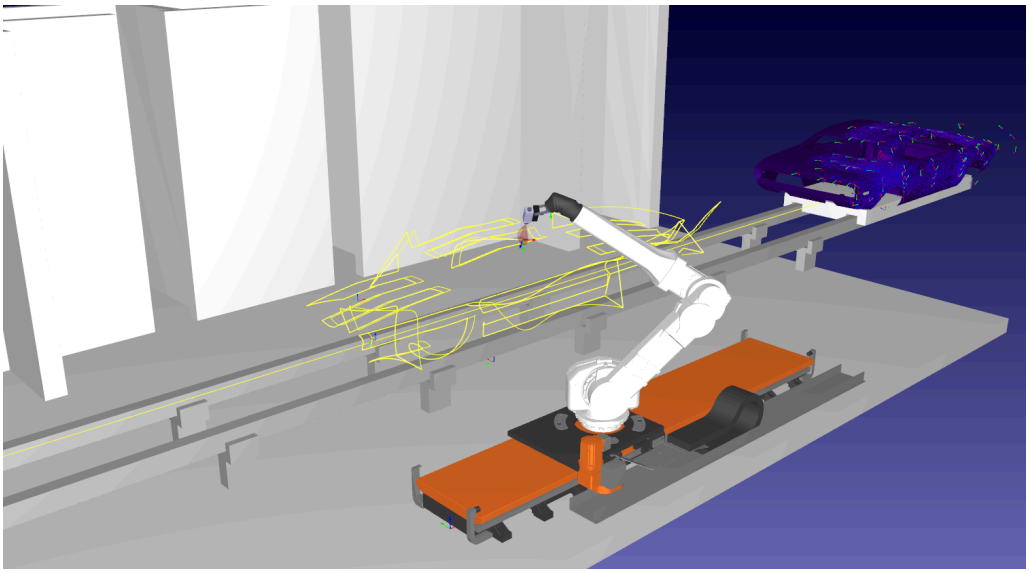
Singularity handling: my first hood program stopped halfway because J5 went from -13 to -7 to +23 degrees, passing through the wrist singularity at J5 = 0. I fixed it by rotating the gun about its own spray axis, which doesn't change the spray, so J5 stays on one side and at least 15 degrees from zero within every pass. Between panels, each program starts at the corner nearest the previous one, and the roof uses the same wrist configuration as the trunk to avoid a large wrist rotation.

6. Collision checking and spray simulation

Robot links are checked against the car, booth, conveyor and track, and every program reports "No collisions detected". Two pairs are excluded: the gun against wrist link 6 (they are bolted together) and the gun against the car (the downloaded gun has a solid decorative spray cone). During conveyor moves the robot parks clear of the car, which I checked over the full conveyor travel.
For the paint I adapted RoboDK's SprayOn macro, with the spray cone sized from the 250 mm standoff and 300 mm width. The spray only registered hits after I split the car into four sections. The car is a free visual model rather than CAD, so its mesh has issues and the paint doesn't stick on a few areas.

7. Results

Result	Value
Panels painted (one side)	Hood, front fender, door, rear fender, trunk, roof
Paint targets	71 across six programs
Painting time, one side	180.6 s
Full cell cycle	About 225 s (conveyor in 15 s, painting 181 s, conveyor out 26.5 s)
Path check	All programs valid, no collisions detected
Transfer efficiency	About 70% of spray particles on the body (simulation estimate)



8. Limitations and next steps

- Only one side is programmed. The second robot would run the same programs mirrored.
- With 3 targets per pass, the gun cuts slightly across curved panels. A production program would place targets every 50 to 100 mm.
- A few strokes are joint moves, mostly near the roof centre where the robot is furthest from the track. A raised track or the real 7-axis IRB 5500-27 would help.
- The conveyor is stop-and-go. Painting a moving car (line tracking) is the next step.
- A clean CAD car body would give better spray results.

9. Links

GitHub: https://github.com/1tsMS/paint-booth-sim
Google Drive: Mohammed Salah Paint Simulation

Car model: "dodge car body" by Dorin.Stoica (Sketchfab), CC-BY-4.0. Robot, track and spray gun models from the RoboDK library.